

Integration Manual

for S32K1_S32M24X WDG Driver

Document Number: IM2WDGASRR21-11 Rev0000R3.0.0 Rev. 1.0

1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	4
2.3 About This Manual	4
2.4 Acronyms and Definitions	5
2.5 Reference List	5
3 Building the driver	6
3.1 Build Options	6
3.1.1 GCC Compiler/Assembler/Linker Options	6
3.1.1.1 GCC Compiler Options	6
3.1.1.2 GCC Assembler Options	8
3.1.1.3 GCC Linker Options	9
3.1.2 IAR Compiler/Assembler/Linker Options	9
3.1.2.1 IAR Compiler Options	9
3.1.2.2 IAR Assembler Options	10
3.1.2.3 IAR Linker Options	11
3.1.3 GHS Compiler/Assembler/Linker Options	11
3.1.3.1 GHS Compiler Options	11
3.1.3.2 GHS Assembler Options	13
3.1.3.3 GHS Linker Options	13
3.2 Files required for compilation	14
3.3 Setting up the plugins	18
4 Function calls to module	21
4.1 Function Calls during Start-up	21
4.2 Function Calls during Shutdown	21
4.3 Function Calls during Wake-up	21
5 Module requirements	22
5.1 Exclusive areas to be defined in BSW scheduler	22
5.2 Exclusive areas not available on this platform	25
5.3 Peripheral Hardware Requirements	25
5.4 ISR to configure within AutosarOS - dependencies	26
5.5 ISR Macro	26
5.5.1 Without an Operating System	26
5.5.1.1 Using Software Vector Mode	26
5.5.1.2 Using Hardware Vector Mode	26
5.5.2 With an Operating System	27
5.6 Other AUTOSAR modules - Dependencies	27
5.7 Data Cache Restrictions	27
5.8 User Mode support	27
5.8.1 User Mode configuration in the module	27
5.8.2 User Mode configuration in AutosarOS	28
5.9 MultiPartition support	28
6 Main API Requirements	30
6.1 Main function calls within BSW scheduler	30
6.2 API Requirements	30
6.3 Calls to Notification Functions, Callbacks, Callouts	30
7 Memory allocation	31
7.1 Sections to be defined in <code>__driver__MemMap.h</code>	31



7.2 Linker command file	32
8 Integration Steps	33
9 External assumptions for driver	34

Chapter 1

Revision History

Revision	Date	Author	Description
1.0	07.03.2025	NXP RTD Team	S32K1_S32M24X Real-Time Drivers AUTOSAR R21-11 Version 3.0.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for the *Wdg* Driver with *S32K1_S32M24X* microcontrollers.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k116_qfn32
- s32k116_lqfp48
- s32k118_lqfp48
- s32k118_lqfp64
- s32k142_lqfp48
- s32k142_lqfp64
- s32k142_lqfp100
- s32k142w_lqfp48
- s32k142w_lqfp64
- s32k144_lqfp48
- s32k144_lqfp64 / MWCT1014S_lqfp64
- s32k144_lqfp100 / MWCT1014S_lqfp100
- s32k144_mapbga100
- s32k144w_lqfp48
- s32k144w_lqfp64
- s32k146_lqfp64

- s32k146_lqfp100 / MWCT1015S_lqfp100
- s32k146_mapbga100 / MWCT1015S_mapbga100
- s32k146_lqfp144
- s32k148_lqfp100
- s32k148_mapbga100 / MWCT1016S_mapbga100
- s32k148_lqfp144
- s32k148_lqfp176
- s32m241_lqfp64
- s32m242_lqfp64
- s32m243_lqfp64
- s32m244_lqfp64

All of the above microcontroller devices are collectively named as S32K1_S32M24X. Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language
EWM	External Watchdog Monitor
WDOG	Watchdog Timer
AEWDOG	Alive watchdog

2.5 Reference List

#	Title	Version
1	Specification of Driver	AUTOSAR Release R21-11
2	S32K1xx Reference Manual	Rev.14.1, 01/2024
3	S32M24x Reference Manual	Rev. 5, 12/2024
4	S32K1xx Data Sheet	Rev. 14, 08/2021
5	S32M2xx Data Sheet	Rev. 7, 12/2024
6	S32K116 Errata	Mask Set Errata for Mask 0N96V Rev. 17/JAN/2024, 1/2024
7	S32K118 Errata	Mask Set Errata for Mask 0N97V Rev. 15/JAN/2024, 1/2024
8	S32K142 Errata	Mask Set Errata for Mask 0N33V Rev. 15/JAN/2024, 1/2024
9	S32K144 Errata	Mask Set Errata for Mask 0N57U Rev. 15/JAN/2024, 1/2024
10	S32K144W Errata	Mask Set Errata for Mask 0P64A Rev. 03/JUL/2024, 7/2024
11	S32K146 Errata	Mask Set Errata for Mask 0N73V Rev. 15/JAN/2024, 1/2024
12	S32K148 Errata	Mask Set Errata for Mask 0N20V Rev. 15/JAN/2024, 1/2024
13	S32M242 Errata	Mask Set Errata for Mask N33V+P69K Rev.1, 8/2024
14	S32M244 Errata	Mask Set Errata for Mask P64A+P69K Rev.1, 8/2024

Chapter 3

Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [IAR Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 10.2.0 20200723 (Build 1728 Revision g5963bc8)
- IAR ANSI C/C++ Compiler V8.40.3.228/W32 for ARM Functional Safety
- Green Hills Multi 7.1.6d / Compiler 2020.1.4

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS_T40D2M30I0R0 part of the plugin name is composed as follows:

- T = Target_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative_Id (e.g. D2 identifies S32K1 platform)
- M = SW_Version_Major and SW_Version_Minor
- I = SW_Version_Patch
- R = Reserved

3.1.1 GCC Compiler/Assembler/Linker Options

3.1.1.1 GCC Compiler Options

Compiler Option	Description
-mcpu=cortex-m4	Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)

Compiler Option	Description
-mcpu=cortex-m0plus	Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)
-mthumb	Generates code that executes in Thumb state
-mlittle-endian	Generate code for a processor running in little-endian mode
-mfpv4-sp-d16	Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)
-mfpv4-sp-d16	Specifies the floating-point hardware available on the target (for S32K11x devices)
-mfloat-abi=soft	Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)
-std=c99	Specifies the ISO C99 base standard
-Os	Optimize for size. Enables all -O2 optimizations except those that often increase code size
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-Wall	Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros
-Wextra	This enables some extra warning flags that are not enabled by -Wall
-pedantic	Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option
-Wstrict-prototypes	Warn if a function is declared or defined without specifying the argument types
-Wundef	Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero
-Wunused	Warn whenever a function, variable, label, value, macro is unused
-Werror=implicit-function-declaration	Make the specified warning into an error. This option throws an error when a function is used before being declared
-Wsign-compare	Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.
-Wdouble-promotion	Give a warning when a value of type float is implicitly promoted to double
-fno-short-enums	Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.
-funsigned-char	Let the type char be unsigned by default, when the declaration does not use either signed or unsigned
-funsigned-bitfields	Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned

Compiler Option	Description
-fomit-frame-pointer	Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.
-fno-common	Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit
-fstack-usage	This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis
-fdump-ipa-all	This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps
-c	Stop after assembly and produce an object file for each source file
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32↔M24x devices)
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT↔ as a macro, with definition 1. Allows drivers to be configured in user mode.
-sysroot=	Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib
-specs=nano.specs	Use Newlib nano specs
-specs=nosys.specs	Do not use printf/scanf

3.1.1.2 GCC Assembler Options

Assembler Option	Description
-Xassembler-with-cpp	Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)
-mcpu=cortex-m4	Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)
-mcpu=cortex-m0plus	Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)
-mfpu=fpv4-sp-d16	Specifies the floating-point hardware available on the target (for S32K14x devices)
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x devices)
-mfpu=auto	Specifies the floating-point hardware available on the target (for S32K11x devices)

Assembler Option	Description
-mfloat-abi=soft	Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)
-mthumb	Generates code that executes in Thumb state
-c	Stop after assembly and produce an object file for each source file

3.1.1.3 GCC Linker Options

Linker Option	Description
-Wl,-Map,filename	Produces a map file
-T linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
-entry=Reset_Handler	Specifies that the program entry point is Reset_Handler
-nostartfiles	Do not use the standard system startup files when linking
-mcpu=cortex-m4	Targeted ARM processor for which GCC should tune the performance of the code (for S32K14x or S32M24x devices)
-mcpu=cortex-m0plus	Targeted ARM processor for which GCC should tune the performance of the code (for S32K11x devices)
-mthumb	Generates code that executes in Thumb state
-mfpu=fpv4-sp-d16	Specifies the floating-point hardware available on the target (for S32K14x or S32M24x devices)
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions (for S32K14x or S32M24x devices)
-mfpu=auto	Specifies the floating-point hardware available on the target (for S32K11x devices)
-mfloat-abi=soft	Specifies the floating-point ABI to use. Specifying "soft" causes GCC to generate output containing library calls for floating-point operations (for S32K11x devices)
-mlittle-endian	Generate code for a processor running in little-endian mode
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-lc	Link with the C library
-lm	Link with the Math library
-lgcc	Link with the GCC library
-n	Turn off page alignment of sections, and disable linking against shared libraries
-sysroot=	Specifies the path to the sysroot, for Cortex-M7 it is /arm-none-eabi/newlib
-specs=nano.specs	Use Newlib nano specs
-specs=nosys.specs	Do not use printf/scanf

3.1.2 IAR Compiler/Assembler/Linker Options

3.1.2.1 IAR Compiler Options

Compiler Option	Description
-cpu=Cortex-M4	Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)

Compiler Option	Description
-cpu=Cortex-M0+	Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)
-cpu_mode=thumb	Generates code that executes in Thumb state
-endian=little	Generate code for a processor running in little-endian mode
-fpu=FPv4-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices)
-fpu=none	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)
-e	Enables all IAR C language extensions
-Ohz	Optimize for size. The compiler will emit AEABI attributes indicating the requested optimization goal. This information can be used by the linker to select smaller or faster variants of DLIB library functions
-debug	Makes the compiler include debugging information in the object modules. Including debug information will make the object files larger
-no_clustering	Disables static clustering optimizations. Static and global variables defined within the same module will not be arranged so that variables that are accessed in the same function are close to each other
-no_mem_idioms	Makes the compiler not optimize certain memory access patterns
-no_explicit_zero_opt	Do not treat explicit initializations to zero of static variables as zero initializations
-require_prototypes	Force the compiler to verify that all functions have proper prototypes. Generates an error otherwise
-no_wrap_diagnostics	Does not wrap long lines in diagnostic messages
-diag_suppress=Pa050	Suppresses diagnostic message Pa050
-DIAR	Predefine IAR as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode.
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode.

3.1.2.2 IAR Assembler Options

Assembler Option	Description
-cpu=Cortex-M4	Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)
-cpu=Cortex-M0+	Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)
-fpu=FPv4-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x devices)
-fpu=none	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)
-cpu_mode thumb	Selects the thumb mode for the assembler directive CODE
-g	Disables the automatic search for system include files
-r	Generates debug information

3.1.2.3 IAR Linker Options

Linker Option	Description
-map filename	Produces a map file
-config linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
-cpu=Cortex-M4	Targeted ARM processor for which IAR should tune the performance of the code (for S32K14x or S32M24x devices)
-cpu=Cortex-M0+	Targeted ARM processor for which IAR should tune the performance of the code (for S32K11x devices)
-fpu=FPv4-SP	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. (for S32K14x or S32M24x devices)
-fpu=none	Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). No FPU. (for S32K11x devices)
-entry __start	Treats __start as a root symbol and start label
-enable_stack_usage	Enables stack usage analysis. If a linker map file is produced, a stack usage chapter is included in the map file
-skip_dynamic_initialization	Dynamic initialization (typically initialization of C++ objects with static storage duration) will not be performed automatically during application startup
-no_wrap_diagnostics	Does not wrap long lines in diagnostic messages

3.1.3 GHS Compiler/Assembler/Linker Options

3.1.3.1 GHS Compiler Options

Compiler Option	Description
-cpu=cortexm4	Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices)
-cpu=cortexm0plus	Selects target processor: Arm Cortex M0+ (for S32K11x devices)
-thumb	Selects generating code that executes in Thumb state

Compiler Option	Description
-fpu=vfpv4_d16	Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x or S32M24x devices)
-fsingle	Use hardware single-precision, software double-precision FP instructions (for S32K14x or S32M24x devices)
-fsoft	Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices)
-C99	Use (strict ISO) C99 standard (without extensions)
-ghstd=last	Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)
-Osize	Optimize for size
-gnu_asm	Enables GNU extended asm syntax support
-dual_debug	Generate DWARF 2.0 debug information
-G	Generate debug information
-keeptempfiles	Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory
-Wimplicit-int	Produce warnings if functions are assumed to return int
-Wshadow	Produce warnings if variables are shadowed
-Wtrigraphs	Produce warnings if trigraphs are detected
-Wundef	Produce a warning if undefined identifiers are used in #if preprocessor statements
-unsigned_chars	Let the type char be unsigned, like unsigned char
-unsigned_fields	Bitfields declared with an integer type are unsigned
-no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup
-no_exceptions	Disables C++ support for exception handling
-no_slash_comment	C++ style // comments are not accepted and generate errors
-prototype_errors	Controls the treatment of functions referenced or called when no prototype has been provided
-incorrect_pragma_warnings	Controls the treatment of valid #pragma directives that use the wrong syntax
-c	Stop after assembly and produce an object file for each source file
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)

Compiler Option	Description
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver (for S32K14x or S32M24x devices)
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT as a macro, with definition 1. Allows drivers to be configured in user mode

3.1.3.2 GHS Assembler Options

Assembler Option	Description
-cpu=cortexm4	Selects target processor: Arm Cortex M4 (for S32K14x or S32M24x devices)
-cpu=cortexm0plus	Selects target processor: Arm Cortex M0+ (for S32K11x devices)
-fpu=vfpv4_d16	Specifies hardware floating-point using the v4 version of the VFP instruction set, with 16 double-precision floating-point registers (for S32K14x devices)
-fsingle	Use hardware single-precision, software double-precision FP instructions (for S32K14x devices)
-fsoft	Specifies software floating-point (SFP) mode. This setting causes your target to use integer registers to hold floating-point data and use library subroutine calls to emulate floating-point operations (for S32K11x devices)
-preprocess_assembly_files	Controls whether assembly files with standard extensions such as .s and .asm are preprocessed
-list	Creates a listing by using the name and directory of the object file with the .lst extension
-c	Stop after assembly and produce an object file for each source file

3.1.3.3 GHS Linker Options

Linker Option	Description
-e Reset_Handler	Make the symbol Reset_Handler be treated as a root symbol and the start label of the application
-T linker_script_file.ld	Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)
-map	Produce a map file
-keepmap	Controls the retention of the map file in the event of a link error
-Mn	Generates a listing of symbols sorted alphabetically/numerically by address
-delete	Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them
-ignore_debug_references	Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers
-Llibrary_path	Points to library_path (the libraries location) for thumb2 to be used for linking
-larch	Link architecture specific library
-lstartup	Link run-time environment startup routines. The source code for the modules in this library is provided in the src/libstartup directory

Linker Option	Description
-lind_sd	Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K14x or S32M24x devices)
-lind_sf	Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library (for S32K11x devices)
-v	Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols
-keep=C40_Ip_AccessCode	Avoid linker remove function C40_Ip_AccessCode from Fls module because it is not referenced explicitly
-nostartfiles	Controls the start files to be linked into the executable

3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the **Wdg** driver for **S32K1 S32M24X** microcontrollers. To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. Only files with the same AUTOSAR major and minor versions can be compiled.

Wdg Files

- ..\Wdg_TS_T40D2M30I0R0\include\AeWdog_Ip.h
- ..\Wdg_TS_T40D2M30I0R0\include\AeWdog_Ip_Types.h
- ..\Wdg_TS_T40D2M30I0R0\include\Ewm_Ip.h
- ..\Wdg_TS_T40D2M30I0R0\include\Ewm_Ip_Types.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_43_Instance0.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_43_Instance1.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_43_Instance2.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_Channel.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_ChannelTypes.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_EnvCfg.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_Ipw.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdg_Ipw_Types.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdog_Ip.h
- ..\Wdg_TS_T40D2M30I0R0\include\Wdog_Ip_Types.h
- ..\Wdg_TS_T40D2M30I0R0\src\AeWdog_Ip.c
- ..\Wdg_TS_T40D2M30I0R0\src\AeWdog_Ip_Irq.c
- ..\Wdg_TS_T40D2M30I0R0\src\Ewm_Ip.c
- ..\Wdg_TS_T40D2M30I0R0\src\Ewm_Ip_Irq.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_43_Instance0.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_43_Instance1.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_43_Instance2.c

- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_Channel.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_Ipw.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdg_Ipw_Irq.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdog_Ip.c
- ..\Wdg_TS_T40D2M30I0R0\src\Wdog_Ip_Irq.c

Wdg Generated Files

- Wdg_43_Instance0_[VariantName]_PBcfg.c (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance1_[VariantName]_PBcfg.c (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance2_[VariantName]_PBcfg.c (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance0_[VariantName]_PBcfg.h (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdg_43_Instance1_[VariantName]_PBcfg.h (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdg_43_Instance2_[VariantName]_PBcfg.h (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdg_43_Instance0_Ipw_[VariantName]_PBcfg.c (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance1_Ipw_[VariantName]_PBcfg.c (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance2_Ipw_[VariantName]_PBcfg.c (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdg_43_Instance0_Ipw_[VariantName]_PBcfg.h (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdg_43_Instance1_Ipw_[VariantName]_PBcfg.h (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdg_43_Instance2_Ipw_[VariantName]_PBcfg.h (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdog_Ip_[VariantName]_PBcfg.c (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.

- Ewm_Ip_[VariantName]_PBcfg.c (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- AeWdog_Ip_[VariantName]_PBcfg.c (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. The file contains the definition of the init pointer for the respective variant.
- Wdog_Ip_[VariantName]_PBcfg.h (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Ewm_Ip_[VariantName]_PBcfg.h (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- AeWdog_Ip_[VariantName]_PBcfg.h (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool. This file will contain the external declaration of the pointer to the configuration structure for the respective configuration.
- Wdog_Ip_Cfg.h (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool.
- Ewm_Ip_Cfg.h (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool.
- AeWdog_Ip_Cfg.h (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdog_Ip_Cfg_Defines.h (For Instance 0) - For driver compilation, this file should be generated by the user using a configuration tool.
- Ewm_Ip_Cfg_Defines.h (For Instance 1) - For driver compilation, this file should be generated by the user using a configuration tool.
- AeWdog_Ip_Cfg_Defines.h (For Instance 2) - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdg_Ipw_Cfg_Defines.h - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdg_Cfg_Defines.h - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdg_Cfg.h - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdg_CfgExt.h - For driver compilation, this file should be generated by the user using a configuration tool.
- Wdg_CfgExt.c - For driver compilation, this file should be generated by the user using a configuration tool.

Files from Base common folder

- ..\BaseNXP_TS_T40D2M30I0R0\include\Compiler.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Compiler_Cfg.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\ComStack_Types.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Wdg_MemMap.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Mcal.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Platform_Types.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Std_Types.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\StandardTypes.h

- ..\BaseNXP_TS_T40D2M30I0R0\include\Reg_eSys.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\Soc_Ips.h
- ..\BaseNXP_TS_T40D2M30I0R0\include\SilRegMacros.h

Files from WdgIf folder

- ..\WdgIf_TS_T40D2M30I0R0\include\WdgIf_Types.h

Files from Dem folder

- ..\Dem_TS_T40D2M30I0R0\include\Dem.h
- ..\Dem_TS_T40D2M30I0R0\generate_PC\include\Dem_IntErrId.h
- ..\Dem_TS_T40D2M30I0R0\include\Dem_Types.h
- ..\Dem_TS_T40D2M30I0R0\include\Dem_stub.h
- ..\Dem_TS_T40D2M30I0R0\src\Dem.c
- ..\Dem_TS_T40D2M30I0R0\src\Dem_stub.c

Files from Gpt folder

- ..\Gpt_TS_T40D2M30I0R0\src\Ftm_Gpt_Ip.c
- ..\Gpt_TS_T40D2M30I0R0\src\Ftm_Gpt_Ip_Irq.c
- ..\Gpt_TS_T40D2M30I0R0\src\Gpt.c
- ..\Gpt_TS_T40D2M30I0R0\src\Gpt_Ipw.c
- ..\Gpt_TS_T40D2M30I0R0\src\LPit_Gpt_Ip.c
- ..\Gpt_TS_T40D2M30I0R0\src\Lptmr_Gpt_Ip.c
- ..\Gpt_TS_T40D2M30I0R0\src\SRtc_Ip.c
- ..\Gpt_TS_T40D2M30I0R0\include\Ftm_Gpt_Ip.h
- ..\Gpt_TS_T40D2M30I0R0\include\Ftm_Gpt_Ip_HwAccess.h
- ..\Gpt_TS_T40D2M30I0R0\include\Ftm_Gpt_Ip_Irq.h
- ..\Gpt_TS_T40D2M30I0R0\include\Ftm_Gpt_Ip_Types.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt_EnvCfg.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt_Ipw.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt_Ipw_Irq.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt_Ipw_Types.h
- ..\Gpt_TS_T40D2M30I0R0\include\Gpt_Irq.h
- ..\Gpt_TS_T40D2M30I0R0\include\LPit_Gpt_Ip.h
- ..\Gpt_TS_T40D2M30I0R0\include\LPit_Gpt_Ip_Types.h
- ..\Gpt_TS_T40D2M30I0R0\include\Lptmr_Gpt_Ip.h
- ..\Gpt_TS_T40D2M30I0R0\include\Lptmr_Gpt_Ip_Types.h
- ..\Gpt_TS_T40D2M30I0R0\include\SRtc_Ip.h
- ..\Gpt_TS_T40D2M30I0R0\include\SRtc_Ip_Types.h

Building the driver

- Gpt_Cfg.c (For PC Variant) - This file should be generated by the user using a configuration tool for compilation.
- Gpt_VS_[index]_PBcfg.c (For PB Variant, [index] is Variant Indexer) - This file should be generated by the user using a configuration tool for compilation.

Files from Ae folder

- ..\Ae_TS_T40D2M30I0R0\src\Ae.c
- ..\Ae_TS_T40D2M30I0R0\src\Ae_Ipw.c
- ..\Ae_TS_T40D2M30I0R0\src\Aec_Ip.c
- ..\Ae_TS_T40D2M30I0R0\src\Aec_Ip_Hw_Access.c
- ..\Ae_TS_T40D2M30I0R0\include\Ae.h
- ..\Ae_TS_T40D2M30I0R0\include\Ae_Ipw.h
- ..\Ae_TS_T40D2M30I0R0\include\Aec_Ip.h
- ..\Ae_TS_T40D2M30I0R0\include\Aec_Ip_Hw_Access.h
- ..\Ae_TS_T40D2M30I0R0\include\Aec_Ip_Types.h
- ..\Ae_TS_T40D2M30I0R0\generate_PC\include\Ae_Cfg.h

Files from Det folder

- ..\Det_TS_T40D2M30I0R0\include\Det.h
- ..\Det_TS_T40D2M30I0R0\src\Det.c

Files from Os folder

- ..\Os_TS_T40D2M30I0R0\src\Os_counter_api.c
- ..\Os_TS_T40D2M30I0R0\include\Os.h
- ..\Os_TS_T40D2M30I0R0\include\Os_counter_api.h
- ..\Os_TS_T40D2M30I0R0\include\Os_counter_types.h
- ..\Os_TS_T40D2M30I0R0\include\Os_types_basic.h
- ..\Os_TS_T40D2M30I0R0\include\Os_types_common_public.h
- ..\Os_TS_T40D2M30I0R0\include\Os_types_public.h
- ..\Os_TS_T40D2M30I0R0\include\Os_version.h
- ..\Os_TS_T40D2M30I0R0\generate_PC\include\Os_cfg.h

Files from Platform folder

- ..\Platform_TS_T40D2M30I0R0\src\Platform.c
- ..\Platform_TS_T40D2M30I0R0\include\Platform.h

3.3 Setting up the plugins

The Wdg driver was designed to be configured by using the EB Tresos Studio (version EB Tresos Studio 29.0.0 or later.)

Location of various files inside the WDG module folder

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:

- ..\Wdg_TS_T40D2M30I0R0\config\Wdg.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
 - ..\Wdg_TS_T40D2M30I0R0\autosar\Wdg_<subderivative_name>.epd
- Code Generation Templates for parameters without variation points:
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\AeWdog_Ip_Cfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\AeWdog_Ip_Cfg_Defines.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\AeWdog_Ip_DeviceRegisters.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Ewm_Ip_Cfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Ewm_Ip_Cfg_Defines.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Ewm_Ip_DeviceRegisters.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdg_Cfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdg_CfgExt.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdg_Cfg_Defines.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdg_Ipw_Cfg_Defines.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdog_Ip_Cfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdog_Ip_Cfg_Defines.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\include\Wdog_Ip_DeviceRegisters.h
 - ..\Wdg_TS_T40D2M30I0R0\generate\src\Wdg_CfgExt.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PC\src\Wdg_43_Instance0_Cfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PC\src\Wdg_43_Instance1_Cfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PC\src\Wdg_43_Instance2_Cfg.c
- Code Generation Templates for variant aware parameters:
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\AeWdog_Ip_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Ewm_Ip_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance0_Ipw_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance0_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance1_Ipw_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance1_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance2_Ipw_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdg_43_Instance2_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\include\Wdog_Ip_PBcfg.h
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\AeWdog_Ip_PBcfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Ewm_Ip_PBcfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance0_Ipw_PBcfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance0_PBcfg.c
 - ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance1_Ipw_PBcfg.c

Building the driver

- ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance1_PBcfg.c
- ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance2_Ipw_PBcfg.c
- ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdg_43_Instance2_PBcfg.c
- ..\Wdg_TS_T40D2M30I0R0\generate_PB\src\Wdog_Ip_PBcfg.c

Steps to generate the configuration

1. Copy the below module folders into the Tresos plugins folder.
 - Wdg_TS_T40D2M30I0R0
 - Dem_TS_T40D2M30I0R0
 - BaseNXP_TS_T40D2M30I0R0
 - Resource_TS_T40D2M30I0R0
 - EcuM_TS_T40D2M30I0R0
 - Gpt_TS_T40D2M30I0R0
 - Mcl_TS_T40D2M30I0R0
 - Mcu_TS_T40D2M30I0R0
 - Os_TS_T40D2M30I0R0
 - Ae_TS_T40D2M30I0R0
 - Spi_TS_T40D2M30I0R0
 - Platform_TS_T40D2M30I0R0
2. Set the desired Tresos Output location folder for the generated sources and header files.
3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
4. Generate the configuration files.

Chapter 4

Function calls to module

- [Function Calls during Start-up](#)
- [Function Calls during Shutdown](#)
- [Function Calls during Wake-up](#)

4.1 Function Calls during Start-up

Wdg shall be initialized during STARTUP phase of EcuM initialization. The API to be called for this is Wdg_Init(). The MCU and Gpt module should be initialized before the Wdg is initialized.

Note: If there are multiple WDG hardware instances used on the platform, the API names will expand according to AUTOSAR requirement SRS_BSW_00347. For example, if there are instances 0 and 1 available on the hardware, then the name of the init functions will be Wdg_43_Instance0_Init and Wdg_43_Instance1_Init instead of Wdg_Init().

4.2 Function Calls during Shutdown

None.

4.3 Function Calls during Wake-up

None.

Chapter 5

Module requirements

- Exclusive areas to be defined in BSW scheduler
- Exclusive areas not available on this platform
- Peripheral Hardware Requirements
- ISR to configure within AutosarOS - dependencies
- ISR Macro
- Other AUTOSAR modules - Dependencies
- Data Cache Restrictions
- User Mode support
- MultiPartition support

5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, WDG is using the services of Schedule Manager (SchM) for entering and exiting the exclusive areas. The following critical regions are used in the WDG driver:

WDG_EXCLUSIVE_AREA_00 is used in function `Wdg_Cbk_GptNotification0` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_00 is used in function `Wdg_Cbk_GptNotification1` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_00 is used in function `Wdg_Cbk_GptNotification2` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_01 is used in function `Wdg_43_Instance0_Init` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_01 is used in function `Wdg_43_Instance1_Init` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_01 is used in function `Wdg_43_Instance2_Init` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_02 is used in function `Wdg_43_Instance0_Init` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_02 is used in function `Wdg_43_Instance1_Init` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_02 is used in function `Wdg_43_Instance2_Init` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_03 is used in function `Wdg_43_Instance0_SetMode` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_03 is used in function `Wdg_43_Instance1_SetMode` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_03 is used in function `Wdg_43_Instance2_SetMode` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_04 is used in function `Wdg_43_Instance0_SetMode` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_04 is used in function `Wdg_43_Instance1_SetMode` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_04 is used in function `Wdg_43_Instance2_SetMode` to protect the updates for the global array:

- `Wdg_aePreviousMode[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_05 is used in function `Wdg_43_Instance0_SetTriggerCondition` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_05 is used in function `Wdg_43_Instance1_SetTriggerCondition` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_05 is used in function `Wdg_43_Instance2_SetTriggerCondition` to protect the updates for the global array:

- `Wdg_au64Timeout[Wdg_Instance]`

WDG_EXCLUSIVE_AREA_06 is used in function `Wdg_43_Instance0_Init` to protect the updates for the global array:

Module requirements

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance1_Init to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance2_Init to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance0_SetMode to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance1_SetMode to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance2_SetMode to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance0_Service to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance1_Service to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_06 is used in function Wdg_43_Instance2_Service to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function Wdg_43_Instance0_Init to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function Wdg_43_Instance1_Init to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function Wdg_43_Instance2_Init to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function Wdg_43_Instance0_SetMode to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function Wdg_43_Instance1_SetMode to protect the updates for the global array:

- Wdg_aeStatus[Wdg_Instance]

WDG_EXCLUSIVE_AREA_07 is used in function `Wdg_43_Instance2_SetMode` to protect the updates for the global array:

- `Wdg_aeStatus[Wdg_Instance]`

Exclusive Areas implemented in Low level driver layer (IPL)

Critical Region Exclusive Matrix

Below is the table depicting the exclusivity between different critical region IDs from the WDG driver. If there is an “X” in the table, it means that those 2 critical regions cannot interrupt each other.

Exclusive Area Matrix									
Exclusive Area ID	WDG_EXCLUSIVE_AREA_00	WDG_EXCLUSIVE_AREA_01	WDG_EXCLUSIVE_AREA_02	WDG_EXCLUSIVE_AREA_03	WDG_EXCLUSIVE_AREA_04	WDG_EXCLUSIVE_AREA_05	WDG_EXCLUSIVE_AREA_06	WDG_EXCLUSIVE_AREA_07	
	X	X		X		X			
	X	X		X		X			
			X		X				
	X	X		X		X			
			X		X				
	X	X		X		X			
							X	X	
							X	X	

Figure 5.1 Critical Region Exclusive Matrix

Note: Extracted table from `RTD_WDG_EXCLUSIVE_AREAS.xlsx`

5.2 Exclusive areas not available on this platform

WDG_EXCLUSIVE_AREA_08 is not available.

5.3 Peripheral Hardware Requirements

The watchdog timer (WDOG) with programmable interrupt response is available in S32K1_S32M24X. There are two user-defined responses to a time-out:

Module requirements

- If a time-out occurs, the WDOG generates an interrupt to the processor core and the WDOG reset will take effect in 128 bus clocks from the interrupt vector fetch. The delay can be increased up to 512 LPO cycles by configuring the RCM_SRIE[DELAY] and RCM_SRIE[WDOG] bits.
- If a time-out occurs, the WDOG generates a system reset and sets the RCM_SRS[WDOG] flag.

In addition to these two modes of operation, the watchdog timer also supports a windowed mode. In this mode, the service sequence must be performed in the last part of the time-out period defined by the window register. The window is open when the down counter is less than the value in the WDOG_WIN register. Outside of this window, service sequence writes that the Watchdog will reset the MCU.

The external watchdog monitor (EWM) is designed to monitor external circuits, as well as the microcontroller software flow. This provides a back-up mechanism to the internal watchdog that resets the microcontroller's CPU and peripherals. The overflow of the watchdog counter must not occur if the software code works well and services the watchdog to re-start the actual counter. The EWM differs from the internal watchdog in that it does not reset the microcontroller's CPU and peripherals. The EWM provides an independent EWM_OUT_b signal that when asserted resets or places an external circuit into a safe mode. The EWM_OUT_b signal is asserted upon the EWM counter time-out. An optional external input EWM_in is provided to allow additional control of the assertion of EWM_OUT_b signal actual counter.

5.4 ISR to configure within AutosarOS - dependencies

The following ISR's are used by the WDG driver:

ISR Name	HW INT Vector	Observations
ISR(Wdg_Ipw_Isr)	22	None

The following ISR's are used by Wdog_Ip and Ewm_Ip standalone drivers:

ISR Name	HW INT Vector	Observations
ISR(Wdog_Ip_Isr)	22	None
ISR(Ewm_Ip_Isr)	22	None

5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

5.5.1 Without an Operating System

The macro *USING_OS_AUTOSAROS* must not be defined.

5.5.1.1 Using Software Vector Mode

The macro *USE_SW_VECTOR_MODE* must be defined and the ISR macro is defined as: ***#define ISR(IsrName) void IsrName(void)***

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

5.5.1.2 Using Hardware Vector Mode

The macro *USE_SW_VECTOR_MODE* must not be defined and the ISR macro is defined as: ***#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)***

In this case, the drivers' interrupt handlers must also handle the context save and restore.

5.5.2 With an Operating System

Please refer to your OS documentation for description of the ISR macro.

5.6 Other AUTOSAR modules - Dependencies

Watchdog Interface

- This module is necessary for importing the mode's type in which the watchdog can be set up. Configuration dependency to other module WDG Driver Integration Manual Module requirements None.

Dependencies

- Base: The Base module contains the common files/definitions needed by all MCAL modules.
- Dem: The Dem module is used for enabling reporting of production relevant error status. The API function used is `Dem_SetEventStatus()`.
- Det: The Det module is used for enabling Development error detection. The API function used is `Det_ReportError()`. The activation / deactivation of Development error detection is configurable using the 'WdgDevErrorDetect' configuration parameter.
- EcuC: The EcuC module is used for ECU configuration. MCAL modules need EcuC to retrieve the variant information.
- Mcu: The Mcu driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required by other MCAL software modules. The clocks need to be initialized prior to using the Wdg driver.
- Gpt: This module is required to reset periodically the watchdog timeout counter.
- Mcl: Module is needed by Gpt functions.
- Resource: Resource module is used to select microcontroller's derivatives. Current driver has support for the following derivatives, everyone having attached a Resource file: s32m242_lqfp64, s32m244_lqfp64.
- Rte: The RTE module is needed for implementing data consistency of exclusive areas that are used by Wdg module.
- Os: The OS module is used for OS configuration. RTD modules need OS to retrieve the application information.

5.7 Data Cache Restrictions

None.

5.8 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

5.8.1 User Mode configuration in the module

Wdg registers are not protected in the user mode.

The Wdg User Mode setting is read only in the configurator, so the Wdg driver can run in user mode.

5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may have the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_IpTrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_IpTrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

5.9 MultiPartition support

- The WDG implements the "MCAL Distribution" according to type II, in which the mappable element is set to HW Unit. For additional details, please refer to **AUTOSAR_EXP_BSWDistributionGuide**.
- The WDG and the mappable elements can be allocated to one or several ECUC partitions, by means of "WdgEcucPartitionRef". If the WDG is mapped to zero ECUC partitions, the WDG behavior reverts to single-partition implementation, similar to previous Autosar versions. If the WDG is mapped to one or more ECUC partitions, the WDG enforces the following multipartition assumptions:
 - The WDG assumes that there is a single ECUC partition per partition. The module internally uses the Partition ID returned by `Wdg_GetUserID` API to reference the appropriate global data and configuration elements.
 - The WDG assumes the `EcucPartitionIDs` are defined in a compact/consecutive order, starting from zero. The rationale is that the number of `EcucPartitions` is used for dimensioning the WDG internal variables and the `EcucPartitionIDs` are used for indexing those variables. (AR-86601 Zero based and dense IDs for OS-Cores and OS-Applications).
 - The WDG assumes that initialization is performed on each partition, `Wdg_Init()` is called separately for each partition, using a different configuration structure (Type II).
 - The upper layer must ensure that each partition has its own initialization configuration. For example, if `EcucPartition_1` is assigned to PartitionID 1, the `Wdg_Init` function must be called on Partition 1 with the `Wdg_Config_EcucPartition_1` configuration structure.
 - The WDG will check upon each API call if the requested resource is configured to be available on the current partition, if DET error reporting is enabled.

- The WDG requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multipartition context.
- The WDG assumes that the RTE module implements the EXCLUSIVE AREAS to be partition-aware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different partitions already. Thus, implementing the EXCLUSIVE AREAS in a blocking manner (e.g., WDGn-lock) on a multipartition scope might affect the performance of the driver across multiple partitions, even if they access separate HW elements. For single-partition scope, the EXCLUSIVE AREAS retain the same purpose as in previous AUTOSAR implementations.

Chapter 6

Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API Requirements](#)
- [Calls to Notification Functions, Callbacks, Callouts](#)

6.1 Main function calls within BSW scheduler

None.

6.2 API Requirements

None.

6.3 Calls to Notification Functions, Callbacks, Callouts

Call-back Notifications:

In Gpt Triggered Mode, a Gpt callback notification will be configured, which will periodically trigger the Wdg servicing routine. This notification will be named Wdg_Cbk_GptNotificationX, depending on the Wdg instance number "X" used.

User Notification:

The WDG Driver provides a notification that is called whenever the defined time period is over. The notifications can be configured as pointers to user defined functions. If notification is not desired, NULL_PTR shall be configured.

An example of the syntax of this function is as follows:

```
void Wdg_Notification (void)
```

The function has to be implemented by the user.

Chapter 7

Memory allocation

- [Sections to be defined in __driver__MemMap.h](#)
- [Linker command file](#)

7.1 Sections to be defined in __driver__MemMap.h

Section name	Type of section	Description
WDG_START_SEC_CONST_8	Configuration Data	Start of Memory Section for Config Data
WDG_STOP_SEC_CONST_8	Configuration Data	End of Memory Section for Config Data.
WDG_START_SEC_CONST_32	Configuration Data	Start of Memory Section for Config Data.
WDG_STOP_SEC_CONST_32	Configuration Data	End of Memory Section for Config Data.
WDG_START_SEC_CONST_UNSPECIFIED	Configuration Data	Start of Memory Section for Config Data.
WDG_STOP_SEC_CONST_UNSPECIFIED	Configuration Data	End of Memory Section for Config Data.
WDG_START_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start of Memory Section for Config Data.
WDG_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	End of Memory Section for Config Data.
WDG_START_SEC_CODE	Code	Start of memory Section for Code in flash.
WDG_STOP_SEC_CODE	Code	Stop of memory Section for Code in flash.
WDG_START_SEC_RAMCODE	Code	Start of memory Section for Code in ram.
WDG_STOP_SEC_RAMCODE	Code	Stop of memory Section for Code in ram.
WDG_START_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	Used for variables, structures, arrays, when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are located in nocacheable region initialized with values after every reset.
WDG_STOP_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	End of above section.
WDG_START_SEC_VAR_INIT_16_NO_CACHEABLE	Variables	Used for variables which have to be aligned to 16 bit. These variables are located in nocacheable region and initialized with values after every reset.
WDG_STOP_SEC_VAR_INIT_16_NO_CACHEABLE	Variables	End of above section.

Memory allocation

Section name	Type of section	Description
WDG_START_SEC_VAR_CLEARED↔ _UNSPECIFIED	Variables	Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are cleared to zero by start-up code (BBS).
WDG_STOP_SEC_VAR_CLEARED↔ _UNSPECIFIED	Variables	End of above section.
WDG_START_SEC_VAR_CLEARED↔ _UNSPECIFIED_NO_CACHEABLE	Variables	Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are located in nocacheable region and cleared to zero by start-up code (BBS).
WDG_STOP_SEC_VAR_CLEARED↔ _UNSPECIFIED_NO_CACHEABLE	Variables	End of above section.

7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>_MemMap.h.

Chapter 8

Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

Chapter 9

External assumptions for driver

The section presents requirements that must be complied with when integrating the WDG driver into the application.

External Assumption Req ID	External Assumption Text
SWS_Wdg_00144	The Wdg Manager (or other entities) shall control the watchdog driver via a so called trigger condition: as long as the trigger condition is valid the Wdg Driver services the watchdog hardware, if the trigger condition becomes invalid the Wdg Driver stops triggering and the watchdog expires. The semantics of the trigger condition can be interpreted as a "permission to service the watchdog for the next n milliseconds". Within this time frame the trigger condition has to be updated by the controlling entity else the watchdog will expire. Handover of the watchdog control logic is simply done by shared usage of the trigger condition (e.g. during startup / shutdown).
EA_RTD_00053	The application shall ensure that Wdg_SetTriggerCondition() function is not preempting itself or any of the following WDG functions:- Wdg_Init()-Wdg_SetMode()
EA_RTD_00055	If WdgRunArea is set to RAM, then the application shall execute all the code which interacts with WDG from RAM (this means also at least DET, DEM, Gpt, SchM,application code will be executed from RAM). Note: Motivation 1: Except the boot loader use case when entire software may run from RAM, there is no other obvious use case when WDG should run from RAM , especially a use case when WDG should run from RAM and other modules from ROM.
EA_RTD_00056	If WdgRunArea is set to ROM, then the application shall execute all the code which interacts with WDG from ROM (this means also at least DET, DEM, Gpt, SchM,application code will be executed from ROM).
EA_RTD_00071	If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used.
EA_RTD_00079	It shall be the integrator responsibility to configure the additional hardware resources (i.e. FCCU) to handle the signals from watchdog modules to generate either an interrupt or a reset.
EA_RTD_00082	When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: Rationale: This ensures that no other buffers/variables compete for the same cache lines.
EA_RTD_00093	The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero.

External Assumption Req ID	External Assumption Text
EA_RTD_00096	The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used.
EA_RTD_00097	The application, after a successful call of Wdg_ClearResetRequest, shall set the watchdog into an active mode using Wdg_SetMode.
EA_RTD_00106	Standalone IP configuration and HL configuration of the same driver shall be done in the same project
EA_RTD_00107	The integrator shall use the IP interface only for hardware resources that were configured for standalone IP usage. Note: The integrator shall not directly use the IP interface for hardware resources that were allocated to be used in HL context.
EA_RTD_00108	The integrator shall use the IP interface to build a CDD, therefore the BSWMD will not contain reference to the IP interface
EA_RTD_00113	When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention Call<Function_Name>TRUSTED(parameter1,parameter2,...) in Integration/User code. They need to be visible in Os.h for the driver to call them. They will call RTD <Function_Name>() as trusted functions in OS specific manner.
EA_RTD_00126	The integrator shall configure the GPT timer to ensure proper servicing of the WDG within correct window period.

How to Reach Us:

Home Page:

nxp.com

Web Support:

nxp.com/support

Limited warranty and liability - Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes - NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use in automotive applications - This NXP product has been qualified for use in automotive applications. If this product is used by customer in the development of, or for incorporation into, products or services (a) used in safety critical applications or (b) in which failure could lead to death, personal injury, or severe physical or environmental damage (such products and services hereinafter referred to as "Critical Applications"), then customer makes the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, safety, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP. As such, customer assumes all risk related to use of any products in Critical Applications and NXP and its suppliers shall not be liable for any such use by customer. Accordingly, customer will indemnify and hold NXP harmless from any claims, liabilities, damages and associated costs and expenses (including attorneys' fees) that NXP may incur related to customer's incorporation of any product in a Critical Application.

Applications - Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale - NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at <http://www.nxp.com/profile/terms>, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control - This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at <mailto:PSIRT@nxp.com>) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

NXP B.V. - NXP B.V. is not an operating company and it does not distribute or sell products.